EVASPA: EVApotranspiration from SPAce

Overview

Ensemble approach

- Uses multiple contextual algorithms based on EF models
- Computes average and dispersion of results across algorithms
- Provides final EF estimate and reliability assessment

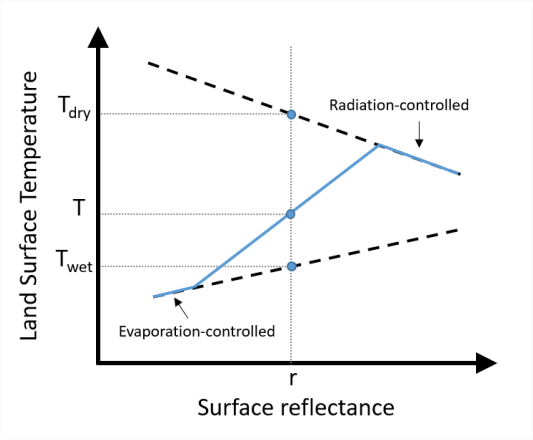
See (Gallego-Elvira et al. 2013)

Methodological principle: Model Assumptions & Limitations

- Requires only incident radiation data (no auxiliary meteorological inputs)
- Assumes land surface temperature variations result from variations in water content, vegetation, or surface albedo
- Assumes no variations in climatic variables, no variations in aerodynamic parameters, and no variations related to topography across the domain
- **Limitation:** Algorithm fails if these uniform conditions aren't met

Methodological principle

- Correlation between surface temperature and reflectance of areas subject to constant atmospheric forcing
- Two distinct thermal regimes exist under constant atmospheric forcing:
  - **Evaporation-controlled** (lower temperature): Temperature changes driven by soil moisture availability
  - **Radiation-controlled** (higher reflectance): Temperature decreases with increasing reflectance when soil moisture is depleted and evaporation ceases



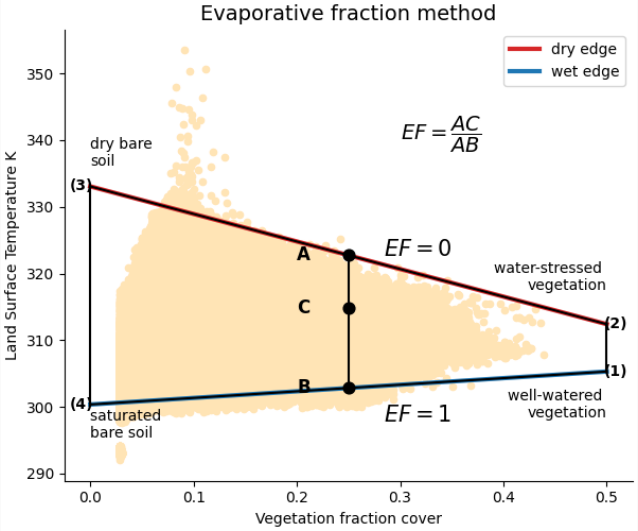
Algorithm: Evaporative fraction

Definition

- Quantity of energy which is allocated to evapotranspiration in regards to the available energy.
- Defined as

$$EF = \frac{LE}{LE + H} = \frac{LE}{R_n - G}$$

Algorithm: Evaporative fraction estimation



$$EF = \frac{T_{dry} - T}{T_{dry} - Twet}$$

- For fully wet conditions, i.e. T_{wet} where $LE = Rn - G$ and $H = 0$
- For fully dry conditions, i.e. T_{dry} where $H = Rn - G$ and $LE = 0$

Algorithm: Ensemble approach

1. Several methods are used to estimate dry and wet edges related to several LST vs variables
2. Each EF is estimated for each methods
3. EF is deduced by merging each EF

Algorithm: Evapotranspiration from evaporative fraction

$$ET = \frac{LE}{L} = \frac{EF(R_n - G)}{L}$$

Gallego-Elvira, Belen, Albert Olioso, Maria Mira, et al. 2013. *EVASPA (EVApotranspiration Assessment from SPAce) Tool: An Overview. Procedia*. January.